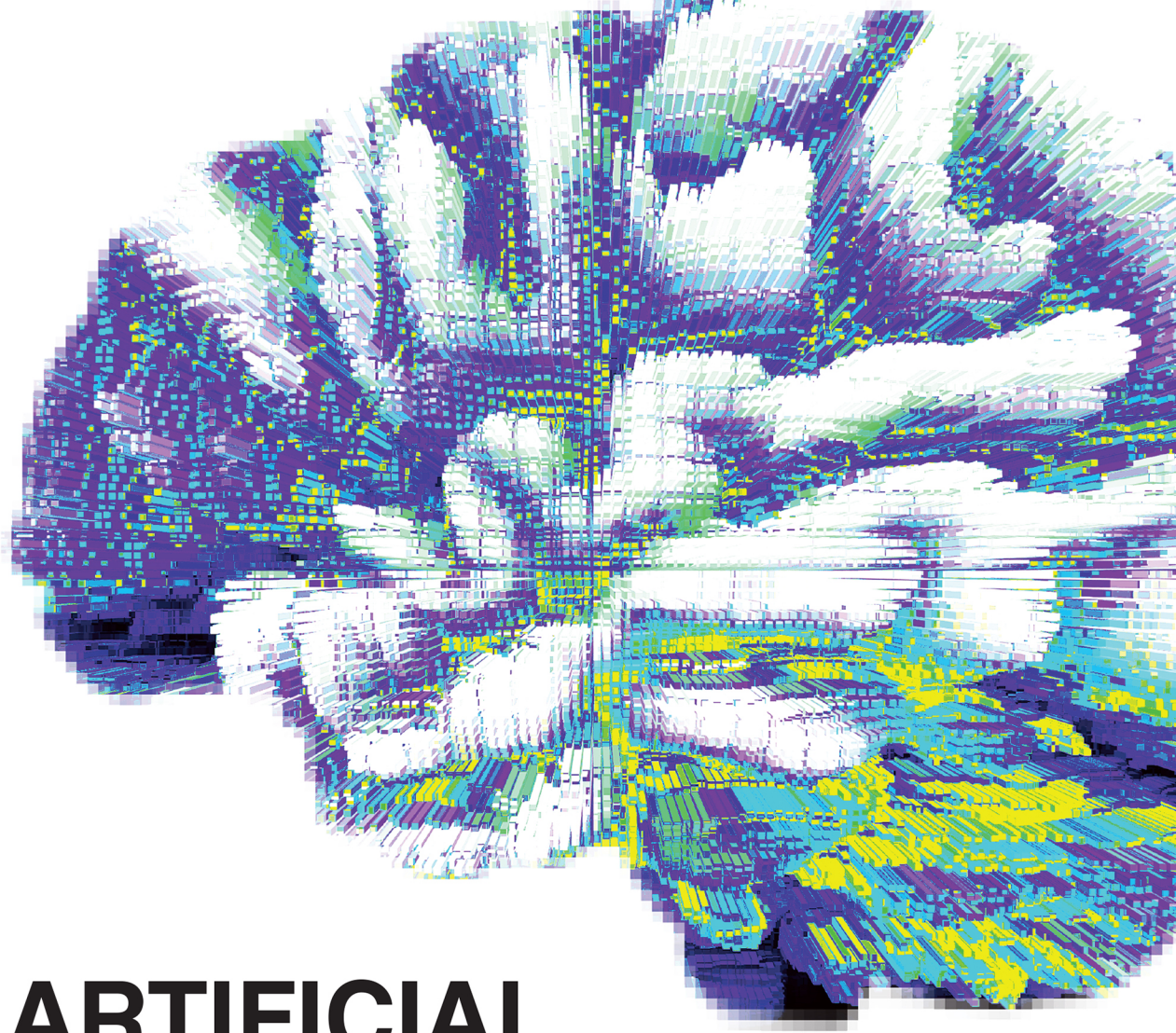




ARTIFICIAL INTELLIGENCE

IN PERFORMANCE-DRIVEN DESIGN

THEORIES, METHODS, AND TOOLS

Edited by

NARJES ABBASABADI • MEHDI ASHAYERI

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Artificial Intelligence in Performance-Driven Design

Theories, Methods, and Tools

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6

Efficient Parametric Design-Space Exploration with Reinforcement Learning-Based Recommenders

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Introduction

Advancements in technology and the emergence of simulation tools have transformed the design process, enabling architects to predict the performance of designs before execution. This shift from intuitive cognition to data-driven design has led to more accurate and confident design solutions. The use of parametric models (Gu, Yu, and Behbahani 2021) further enhances the process by generating multiple design alternatives within a defined design space (Ritter, Geyer, and Borrmann 2013), which encompasses a vast range of options for exploration. By considering the entire design space and analyzing various design alternatives, designers can avoid costly expenses and achieve better performance outcomes in the early stages of the design process.

Exploring the entire design space manually can be overwhelmingly tedious for a design team. Therefore, optimization becomes necessary to identify the best design option (Rahmani Asl et al. 2014). Finding a design solution that satisfies functional requirements and environmental objectives while optimizing efficiency and lowering costs is the aim of optimization. Various algorithms, such as the genetic algorithm, inspired by natural selection, are widely used for architectural optimization (Echenagucia 2014). These algorithms strategically iterate over different solutions, assessing their fitness based on predefined criteria such as energy efficiency and structural stability. However, these optimization processes prioritize quantitative outcomes and may generate solutions that are functionally superior but esthetically or spatially less appealing. This approach neglects the qualitative aspects of design options (Saldana Ochoa et al. 2021). The pure optimization approach is considered an oversimplification of the design problem, as it overlooks the designer's preferences and personal biases. Designers seek to achieve the best possible compromise between their preferences and performance results, which may be difficult with conventional optimization techniques.

Another limitation of existing computational optimization approaches is the time-consuming nature of performance simulations, particularly for computationally intensive tasks. As a result, optimizing a large design space can take hours or even days, consuming

significant time and resources. To speed up the design optimization procedure, surrogate models are being widely used (Mueller and Ochsendorf 2015). These models can rapidly predict a design's performance without requiring a complete simulation for each iteration. A collection of previously assessed designs and their accompanying performance measures are used to train machine learning (ML) models. Once trained, these models can quickly approximate performance outcomes (Wortmann et al. 2015), speeding up the optimization process. However, accurate predictions require a substantial number of simulations to prepare the training set for the surrogate model, especially when dealing with vast design spaces.

Designers face a significant challenge when navigating large design spaces, but researchers have made remarkable progress by incorporating ML algorithms. To gain an overall understanding of the vast design space generated, a study implemented Combinatorial Equilibrium Modeling (CEM) (Ohlbrock and Schwartz 2016) and clustered design options using self-organizing maps (SOM). Visualizing clusters with the assistance of the Uniform Manifold Approximation and Projection algorithm (UMAP) (Fuhriemann et al. 2018) provided insights into the design space. Another study (Saldana Ochoa et al. 2021) followed a similar approach but also sought to incorporate the designer's understanding of non-quantifiable aspects. By including human designers' preferences in their proposed framework, they introduced qualitative limitations alongside the quantitative ones. In the application of the five-step design framework, Saldana Ochoa et al. (2021) utilized it to explore the design of a stadium, which is both structurally sound and looks good, while Bertagna et al. (2021) used the same process to revisit an existing structure. In the case of the existing structure, they generated additional options using CEM and clustered them using SOM to obtain a low-dimensional representation of the design space. The arrangement of design options in a two-dimensional grid proved beneficial for considering multiple criteria such as structural, daylight, and radiance analysis.

There are two main approaches to incorporating user preferences in the multiobjective optimization process. First, users can adjust the weightings between objectives during a multi-objective search. Second, users can choose which designs to consider for each generation (Simons, Parmee, and Gwynllyw 2010). Reeh (2019) achieved the design of a single-family home using a genetic algorithm-based optimization approach that allows users to filter design possibilities after each iteration. The non-dominated sorting genetic algorithm (NSGA) (Mohapatra et al. 2015) allowed users to personalize the optimization process to align with their preferences in decision-making. (Harding and Brandt-Olsen (2018) developed the "Biomorpher" grasshopper plug-in based on a similar approach, grouping similar design options, and presenting them in an interactive dashboard to assist designers in filtering out undesired options. This approach employed interactive evolutionary computation (IEC) (Takagi 2001) to achieve design solutions that satisfy both performance and esthetic preferences. To incorporate the designer's qualitative measures Yi (2019) proposed a framework that derives numerical measurements for esthetic preferences through a survey, where the participants vote for their preferred option in an interactive dashboard created using the Grasshopper plug-in Human UI ("Human UI" 2016).

As researchers delve into innovative methods for design space exploration, they are actively seeking collaborations with diverse experts in built environment studies. Unlocking the potential of design computing and ML is the hope behind these collaborations. While

multiobjective optimization processes have been proposed in previous studies, an unexplored avenue with significant potential lies in utilizing recommender systems (Ricci, Rokach, and Shapira 2022). These systems, akin to those recommending products or media content, can be employed to suggest design options, alleviating users from the intricate complexities of comprehending the entire design space. This approach enables stress-free exploration while aligning with users' intentions.

Reinforcement learning-based recommender systems (Choi et al. 2018) present a pioneering paradigm among various recommender systems utilized on websites. Diverging from conventional collaborative or content-based filtering techniques (Techlabs 2021), which heavily rely on exhaustive user information for effective suggestions, reinforcement learning-based systems offer a compelling solution. Unlike traditional methods that often demand users' personal details or prolonged data collection periods, these systems rapidly grasp user preferences through interactive experiences, addressing the "cold start" challenge (Lika, Kolomvatsos, and Hadjiefthymiades 2014). By leveraging reinforcement learning principles, these systems swiftly acquire accurate insights, ensuring impartial recommendations devoid of biases related to demographics or background. This streamlined yet potent approach has contributed to the surging popularity of reinforcement learning-based recommender systems. Such systems operate ethically, concentrating on providing superior recommendations while steering clear of commercial biases. The integration of this tool into parametric design exploration is anticipated to herald an exciting era of participatory design and uninhibited exploration.

This chapter presents a framework that utilizes AI models to address limitations in the current design process. The framework comprises two subdivisions, which are applied to find a suitable design solution for a four-story office building in Seattle, evaluating annual energy consumption using EnergyPlus. The first subdivision quickly learns performance outcomes through dimensionality reduction and clustering, while the interactive dashboard allows users to provide feedback and receive new design suggestions from a reinforcement learning-based recommender system. This approach enables efficient exploration of the design space with the goal of generating high-performing and visually appealing solutions.

Methodology

In the first section of the study, a design space is created, and ML algorithms are employed to cluster all the design options effectively. This clustering process ensures that buildings with similar performance analyses are grouped together. This step is crucial as it sets the foundation for the second section, where a reinforcement learning-based recommender system is implemented. In the second section, the focus is on the implementation and customization of the reinforcement learning algorithm to achieve a design solution that satisfies both functional objectives and esthetic appeal. A reward function is designed to prioritize high-performing alternatives while incorporating the designer's esthetic preferences. This customization allows the algorithm to guide the designer toward design options that excel in both functional and esthetic aspects. A design dashboard is created to facilitate the exploration of the extensive design space. The reinforcement learning algorithm is trained to understand user preferences through interaction with the dashboard.

Section 01: Clustering Design Options

Design Space Generation

The Design Space comprises 6000 four-storied design options with varying lengths and widths, ranging from 10 to 50 m (see Table 6.1). The building type is a four-storied “mid-rise apartment” (“Commercial Reference Buildings” n.d.) with the ground floor serving as a retail store. The ground floor has the highest window-to-wall ratio (WWR), while the upper floors have different WWRs. The north and south facades of the apartment portion are divided into a 5×3 grid, allowing for a maximum of 5 windows on each floor on each side. The east and west facades are divided into a 3×3 grid, allowing for a maximum of 3 separate windows on each floor on each side. In total, the building can have 48 window segments (see Figure 6.2). The presence of windows in each segment depends on the design option parameters.

A pandas DataFrame (McKinney and others 2011) with 6000 rows and 98 columns is created (see Table 6.1), representing unique design options. The first 25 columns correspond to window positions on the north facade, while columns 15–23, 24–38, and 39–47 represent window positions for the west, south, and east facades, respectively. Each window position is assigned a unique decimal value indicating the WWR and size, with zero, indicating no window and non-zero values ranging between 0.3 and 0.7. Columns 49 and 50 represent the lengths and widths of the design options, while columns 50 through the end define the depth of shades (see Figure 6.1). The DataFrame allows for shade depths of 0 for windows, except for segments with a WWR of zero, where the shade depth is also set to zero to avoid bias.

Table 6.1 Building characteristics for the main Design Space.

Building characteristics	Values
Building width	Range (10—50 m) (east-west)
Building length	Range (10—50 m) (north-south)
Window segments (East-west)	3×3
Window segments (North-South)	5×3
Floor-to-floor height	3 m
Number of floors	4
Shade depth	Range (0–2.5 m) three-sided shade
Window sizes	Range (0–5 m)
Ground floor WWR	0.6
Surrounding context	Not considered
DataFrame configuration	
Number of parameters	98
Amount of data points	6000

North facade 00–14			West facade 15–23			South facade 24–38			East facade 39–47			Length	Width	North shade 50–64			West shade 65–73			South shade 74–88			East shade 89–97		
0	...	14	15	...	23	24	...	38	39	...	47	48	49	50	...	64	65	...	73	74	...	88	89	...	97
32459	0.4	...	0	0	...	0.4	0.3	0	0	0.7	0	0.4	0	0	...	0	0.5	...	0.7	0	...	0	0	...	0
82397	0.7	...	0.4	0	...	0.4	0	0	0.7	0.7	0.3	0	0	0.3	...	0	0	...	0.7	0.3	...	0.6	0.5	...	0.6
1702	0.6	...	0	0	...	0	0.5	0.5	0	0	0.5	0.7	0	0.6	...	0.7	0.6	...	0.6	0	...	0	0	...	0
52330	0	...	0.7	0.6	...	0.7	0	0.3	0	0.5	0	0.6	0.5	0	...	0.4	0.5	...	0	0.3	...	0.4	0	...	0.5
87482	0.6	...	0	0	...	0.5	0	0	0.6	0	0.7	0	0	0	...	0	0	...	0	0	...	0	0	...	0

Figure 6.1 Data structure of the main Design Space.

Table 6.2 Sample data frame of the main Design Space.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
4754	0	0	0.5	0.4	0	0	0.4	0	0	0	0	0.5	0	0.4	0.3	0	0.3	0.3
1091	0.7	0.6	0	0	0.4	0.5	0	0	0.7	0.4	0.7	0	0	0.5	0.5	0.7	0.5	0
2984	0.6	0	0.5	0.5	0.6	0.3	0.3	0.5	0	0.6	0	0	0	0	0.7	0.7	0.7	0
4506	0.7	0.3	0	0.3	0.5	0.6	0	0.3	0	0	0.7	0	0	0	0	0	0	0
4968	0	0	0	0.7	0	0	0.3	0	0	0	0.7	0.7	0.6	0	0	0.5	0.5	0

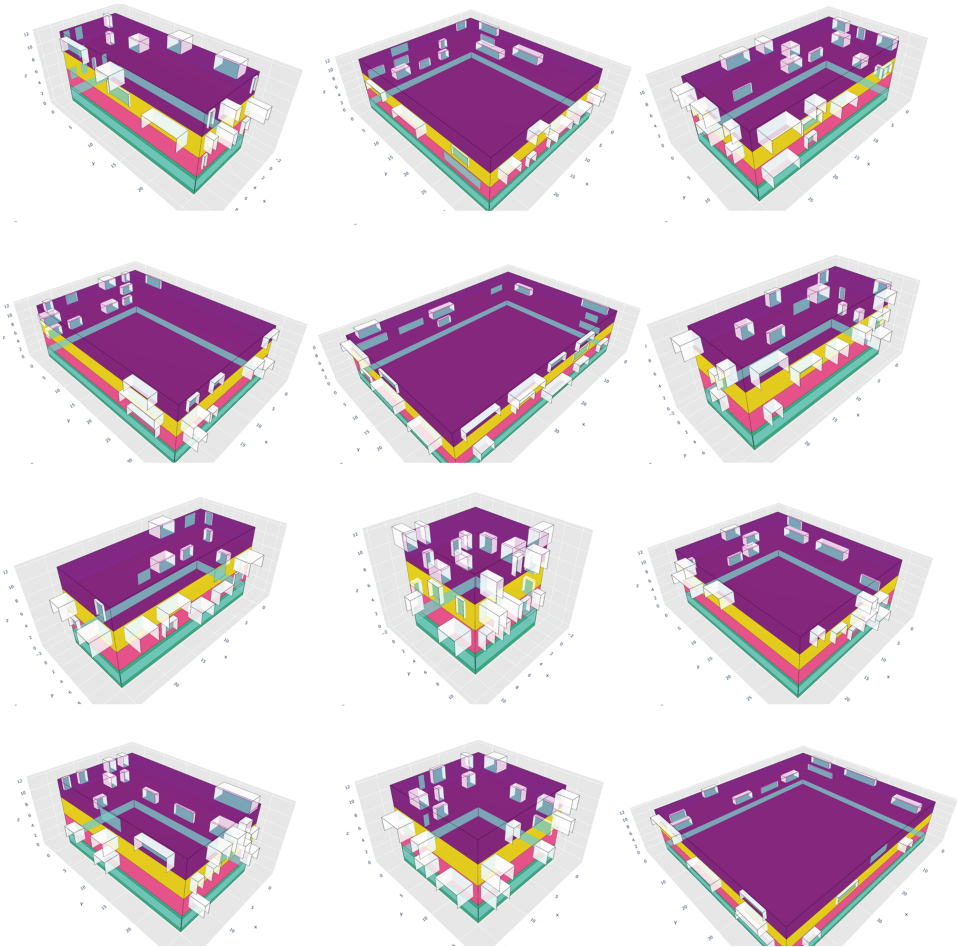


Figure 6.2 Randomly selected 12 design options from the main Design Space.

Performance Evaluation

To recommend design options with lower annual energy consumption, the algorithm needs to know the annual energy consumption of all the 6000 design options in the generated design space. However, simulating all 6000 design options would be extremely time-consuming. To address this issue, the study employs a novel approach where the performance outcome of only 20% of the design options is learned, and this information is used to approximate the performance outcome of all other design options. The performance outcome considered is the annual energy consumption (in kWh) for heating, cooling, and lighting. EnergyPlus (version 22-2-0) is used for the simulations, and the whole process is conducted in Python (version 3.9). The authors utilize an external Python library called “Compas eplus” (Echenagucia [2022] 2023) to communicate with EnergyPlus, developed by the second author of this chapter. The geometries are generated using Python scripts, which are then converted into an EnergyPlus-readable IDF format to generate the outcomes. Table 6.3 provides the properties that were fixed for the building during the simulation which is kept the same for the whole study.

Clustering Design Options

The primary goal is to cluster similar design options to identify the clusters that are preferred by designers during interaction, allowing the algorithm to show more design options from those specific clusters. To understand designers’ preferences, their position in the design space needs to be analyzed, and nearby similar options must be identified for future recommendations. Euclidean distances between different design options can help determine the proximity of data points. The effectiveness of the clustered output depends on the given dataset. In the current dataset, each design has 98 features, resulting in 98-dimensional data points. Having high-dimensional data points with a relatively low number of data instances can lead to the “curse of dimensionality” (Verleysen and François 2005). This phenomenon presents challenges, such as increased computational complexity, decreased model performance, and data sparsity. To address these issues, it is crucial to reduce the dimensions of the dataset before using it for clustering purposes.

Dimensionality Reduction

The study investigated dimensionality reduction methods and found *t*-SNE (*t*-Distributed Stochastic Neighbor Embedding) (van der Maaten and Hinton 2008) ensuring effective

Table 6.3 Simulation configuration followed for all studies.

Simulation configuration	Values
Building type	DOE midrise apartment
Location	Seattle
Wall	Typical Insulated Steel Framed Exterior Wall-R16
Roof	Generic Roof
Floor	Generic Interior Floor
Window	Generic Double Pane
Simulation outcome unit	Kilowatt-hour (kWh)/year

simulation. t-SNE is a widely used nonlinear technique, creating a probability distribution to cluster similar data points and achieve distinct separations between clusters in lower-dimensional space. The Scikit-learn library's TSNE model was used, with hyperparameter adjustments to improve cluster separation (perplexity: 30, early exaggeration: 12). After reducing the data points to three-dimensional representations using the TSNE dimensionality reduction method, the team employed Python's Plotly library (Dabbas 2021) to visualize the outcomes. The reduced dimension data points appeared to be distributed in a large sphere (see Figure 6.3), which posed challenges for density-based clustering methods that were initially considered for grouping the data points. The structure of the data frame contributed to the difficulty in achieving effective clustering. Different parameters have different levels of influence on the performance outcome, which should be informed to the data frame before reducing the dimension.

Adding Weights Method

Scikit-learn Linear regression model (Su, Yan, and Tsai 2012) is utilized to find out the weights of different columns. To examine the accuracy of the weights, the team visualized data points in lower dimensions. By only simulating randomly selected 20% of the data points from the entire design space (1200 design options) and using their performance outcomes to train the linear regression model, we obtained weights that effectively modified the data frame, leading to densified groups of data points. Visualizing these modified data points in a three-dimensional space showed effective clustering (see Figure 6.3).

Now, instead of simulating all the design options, the team only simulated data points near the centroid of the created groups, considering them good representatives of their respective clusters. It was observed that the performance outcomes of other data points in each group deviated very little from the outcome of the data point close to the centroid. Thus, we assumed the performance outcomes of all data points in a cluster to be the same as that of the centroid data point. This assumption resulted in a low “prediction error.”

The mean absolute error (MAE) (Willmott and Matsuura 2005) was calculated for each data point while using this framework, and the percentage of the actual value predicted wrong represented the final MAE. The MAE for all clusters was summed and divided by the number of clusters to calculate the average MAE for the entire clustering method. The average MAE is considered as the “prediction error,” and was found to be 1.1. This means that the performance prediction for any data point can be approximately 98.9% to 101.1% of its actual outcome.

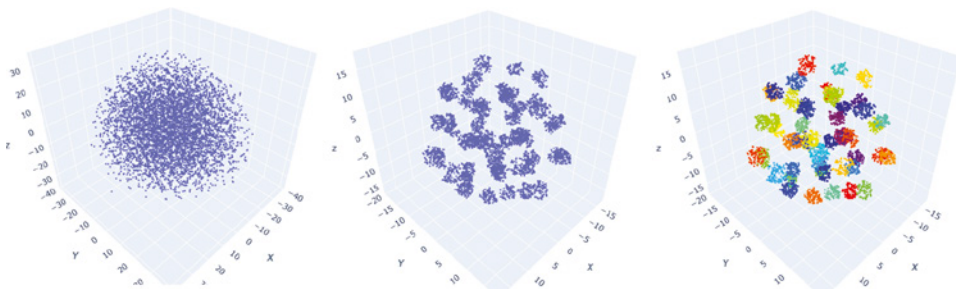


Figure 6.3 Data points visualization in the lower dimension. (Left: without adding any weights; Middle: After adding weights; Right: clusters by Meanshift clustering).

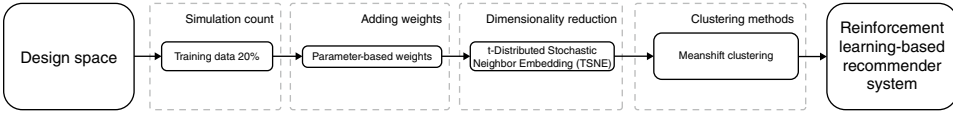


Figure 6.4 Proposed framework.

Clustering Method

In this study, considering that the dimensionality reduction processes aim to densify data points into smaller groups, the Meanshift clustering method was chosen for further investigations (see Figure 6.4). Meanshift is a density-based clustering algorithm known for its ability to identify clusters of varying shapes and sizes (Derpanis 2005). One of its major advantages is the adaptive determination of the number of clusters by adjusting the bandwidth parameter. By manipulating the bandwidth value, the granularity of the clustering results can be controlled, resulting in different numbers of clusters. In this study, a bandwidth value of 0.003 was utilized, which produced a total of 49 clusters (see Figure 6.3).

Section 02: Reinforcement Learning-Based Recommender System

In the second section of the study, the clustered design options and their corresponding performance outcomes are utilized to create a design dashboard using Compas (“COMPAS” n.d.) and Plotly (Dabbas 2021). This design dashboard is accessible online without the need to install any software. The dashboard provides a three-dimensional model of various design options from the design space, allowing designers to view and interact with the designs from different angles.

Process Explanation

Incorporating reinforcement learning (Kaelbling, Littman, and Moore 1996) into the design dashboard allowed us to strategically extract the designer’s preferences. Reinforcement learning is a type of ML where an agent learns from its interactions with the user and makes decisions that align with the user’s preferences. The agent constantly learns from its environment through trial and error, maximizing the rewards it receives for each interaction. The process begins with the agent displaying a random design option to the designer. Three buttons are provided on the dashboard: “like,” “unlike,” and “slide” (see Figure 6.6). If the designer clicks “like,” the cluster to which the design option belongs is rewarded with a positive value. Conversely, if the designer dislikes the option, the reward is a negative value. Sliding the option results in a reward of zero.

The rewards received are used to update the q-value (also known as the action-value function) of each cluster. Each cluster, along with all designs within that cluster, has a preassigned q-value before any interactions begin. These q-values are inversely mapped to the performance outcomes, indicating that higher annual energy consumption leads to lower q-values. The performance outcomes are mapped from 100 to 0 for this system, where a cluster with the highest annual energy consumption per square meter has a q-value of 0, and the cluster with the lowest annual energy consumption per square meter has a value of 100. After each interaction, the q-values are updated based on the user’s feedback, enabling the system to adapt and better align with the designer’s preferences.

Reward Function

The quantitative value of the user's feedback is denoted as the reward (R). A function is established to determine the appropriate reward, which can be positive or negative, for the system based on the user's interaction with the displayed design option. This function takes into account two constants, alpha (α) and gamma (γ), along with the reward (R) and the previous q-value. The old q-value is then updated with the new value obtained from this reward function (see Equation 1). This update equation is known as the Bellman equation (Barron and Ishii 1989), a fundamental concept in reinforcement learning. The study simplifies the Bellman equation by presenting only one design option in each iteration, resulting in a single state to consider. If multiple options were displayed simultaneously, the algorithm would need to calculate probabilities based on the q-values to estimate user preferences, which is not explored in this chapter.

Bellman equation for incorporated reinforcement learning

$$f(r) = (1 - \alpha) \times Q + \alpha \times (R + \gamma \times Q) \quad (6.1)$$

α = learning rate

γ = discount factor

R = reward defined by β

Q = Previous q-value

Constants/Hyperparameters

The exploration process in the design dashboard involves four predefined constants that allow designers to customize their experience strategically:

Learning Rate (alpha (α)) This hyperparameter represents the learning rate, which controls the impact of recent interactions on the new q-value compared to past interactions. A higher alpha value gives more weight to recent preferences, allowing designers to adapt and change their preferences over time. Designers need to define the alpha value before starting the exploration process, and it typically ranges between 0 and 1.

Discount Factor (gamma (γ)) The gamma (γ) value is included to facilitate multistate reinforcement learning in the future, although it may not have a significant impact on single-state reinforcement learning. Gamma determines the algorithm's speed in aiming for options with greater q-values. It ensures that designers do not get stuck in a specific set of design options where their q-values may decrease over time. Like the learning rate, gamma also takes a value between 0 and 1, and designers must assign it before the exploration process begins.

Exploration Exploitation Trade-off Epsilon (ϵ) When the user provides feedback, the agent has two options: it can either randomly select a cluster to show design options or choose the cluster with the highest q-value. To prevent excessive jumping across the design space, the system restricts the search to nearby clusters. The epsilon (ϵ) constant determines the likelihood of random exploration. In the Python script, an if function is used to compare a randomly generated number between 0 and 1 with epsilon (ϵ). If the random number is less

than epsilon, the system selects a random nearby cluster. Higher epsilon (ϵ) values increase the chance of random selection, while lower values favor the exploitation of preferred options. The value of epsilon can be adapted over time, allowing users to have less exploration as they gain more experience with the design space.

Priority Trade-off (beta (β)) The constant beta (β) in the reinforcement learning process determines how much the agent prioritizes the user's preferences over the preassigned q-values. It is a value between 0 and 1, where a higher beta value means that the user's preferences have more influence on the agent's decisions. In other words, if the beta is higher, the user's feedback of "like" for a design option will have a greater impact on the agent's recommendation for future designs. This aspect sets the proposed reinforcement learning-based recommender system apart from traditional systems. In conventional algorithms, the main focus is to increase the number of "likes" for individual design options. The suggested system, however, goes beyond that objective. It tries to advise designers on both preferred design alternatives and clusters with high performance.

Reward Alteration with Alpha (α) and Gamma (γ) The reward's magnitude after a positive interaction affects the number of hits required to suppress the highest-performing design option. Determining the perfect reward value for exploration depends on the interrelationship between beta (β) and other constants. The reward function considers all four constants (alpha, gamma, epsilon, and beta) to strike a balance and guide the design exploration process, considering trade-offs. Understanding this interplay is essential in deriving an effective reward function for personalized design exploration. Figure 6.5 illustrates this relationship. The team conducted an investigation to examine how the assigned rewards relate to the number of iterations required to surpass the highest q-value, considering different values of alpha (α) and gamma (γ). It was observed that lower initial reward values result in more iterations for a cluster with the lowest q-value to surpass the highest q-value. It was found that the reward should be greater than 100 to avoid an almost infinite number of iterations for a cluster with a q-value of 0 to surpass a q-value of 100. To account for the possibility that the cluster with the lowest q-value could be the one designer prefer, Beta (β) is employed to specify a suitable reward(R) value for the system, eliminating the need for designers to repeatedly click "like" to express their preferences.

Function Determines beta(β) A function has been created to construct a Pareto frontier, which maps reward values (varying from 100 to 200) on the x-axis of the graph to the number of "likes" necessary to attain a q-value of one hundred of the cluster with the least q-value on the y-axis. A Pareto Frontier is a distribution of optimal solutions, which helps to understand the trade-off between multiple objectives. Here the team focuses to investigate the way to compromise the number of iterations for different reward values. The function utilizes the same Bellman equation used in the algorithm to derive the number of "likes" needed (Eq. 6.1). The exploration factor (epsilon, ϵ) is set to 0 to eliminate exploration effects and solely focus on the frontier.

The function can construct the Pareto frontier with various combinations of alpha (α) and gamma (γ) values. It runs before the exploration process begins, using preassigned values of alpha, gamma, and beta. Beta (β) is important in establishing the amount of "likes" required to give preference to clusters with moderate or reduced q-values, as well as in defining an acceptable reward function. To illustrate, let's discuss a system with gamma 0.01 and alpha

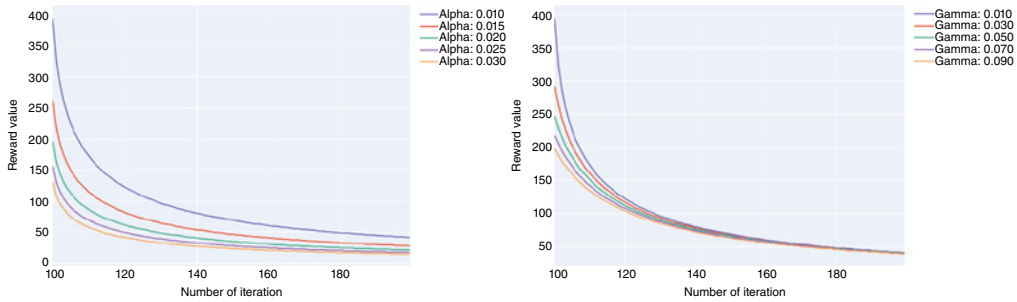


Figure 6.5 Changes in reward value with the number of interactions for different alpha(right) and gamma(left) values.

0.01. The function is used to calculate the number of “likes” for reward values ranging from 100 to 200, resulting in a range of 41 to 395. By multiplying this range (354) by beta (0.6), we obtain the desired number of “likes” (nearly 212). The corresponding reward value can then be determined from the Pareto frontier.

Design Dashboard

The team opted for Plotly, a Python library (Dabbas 2021) with integration into Dash (Hossain et al. 2019), instead of the commonly used Human UI plugin for creating the dashboard. Unlike the dashboard created using the Human UI plugin, which is limited to access within Grasshopper, the dashboard produced by Plotly is accessible via any web browser. Moreover, the Plotly dashboard offers enhanced integration options and greater flexibility in its functionality. The dashboard features a 3D model of the design option, interactive buttons for user input, a line chart illustrating performance changes in kWh, and pixelated images of selected design options. Eight scatter plots below the 3D perspective help visualize changes after each interaction, with the first row showing individual data points and the second row representing clusters as single data points (see Figure 6.6).

- 1) **First row, first column chart:** In order to determine if closely aligned data points are clustered together, this chart colors data points according to their cluster number.
- 2) **First row, second column chart:** The changes in q-values after each repetition are seen in this graph. It excludes the values that were initially allocated and simply shows the changes in the q-values. The q-value for every data point in a cluster is the same.
- 3) **First row, third column chart:** This graph displays the actual performance results of individual data points (annual energy consumption per square meter (kWh)). No remapping was done.
- 4) **First row, fourth column chart:** This chart highlights the displayed data point by marking it black and keeping the rest of it white. By clicking on any data point, a pixelated image of the selected design option appears, facilitating exploration in the lower-dimensional space.
- 5) **Second row, first column chart:** The standard deviation of the average performance outcome across all clusters is color mapped in this graph. It shows which subset of the data has performance outcomes that are substantially different from the mean.

Reinforcement learning based recommender system

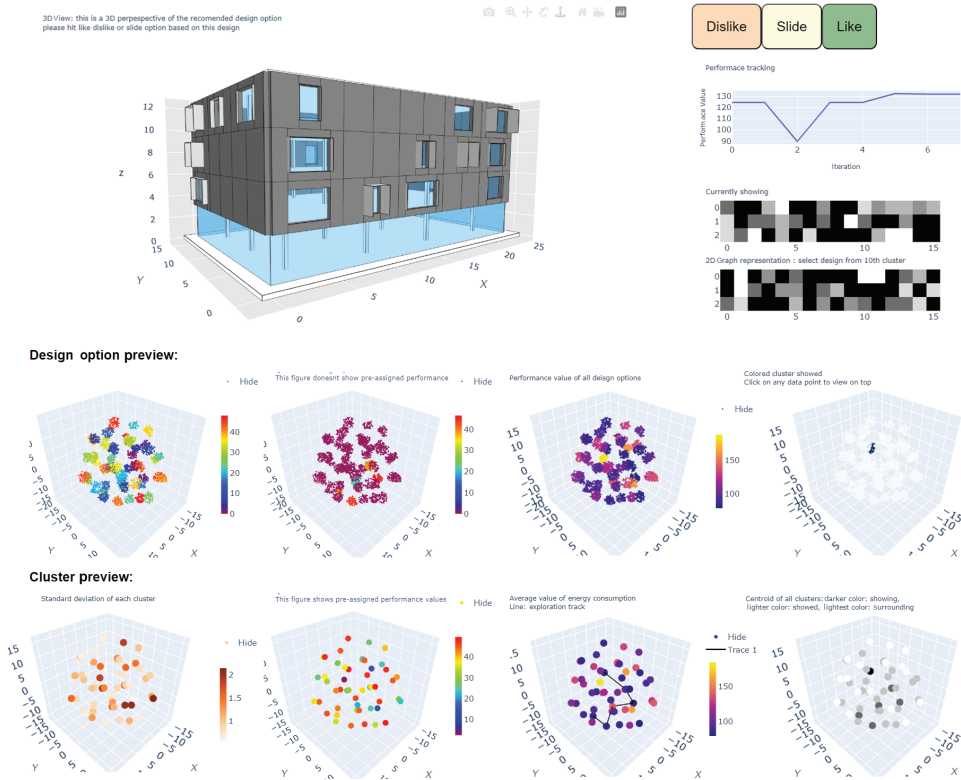


Figure 6.6 Design Dashboard (after 7 iterations).

- 6) **Second row, second column chart:** Showing preassigned q-values for each cluster, this graph reveals the discrepancy between assigned q-values and average performance outcomes.
- 7) **Second row, third column chart:** This graph contains a line that represents maneuvering in the design space after the last interaction and shows the average yearly energy usage per square meter for each cluster. Proper reinforcement learning functioning is indicated if the line consistently moves toward warmer-colored clusters.
- 8) **Second row, fourth column chart:** The current design option's origin cluster is marked in black, whereas previously visited clusters are tinted in gray. It is possible to trace the exploration of the design space and comprehend the algorithm's success by looking at nearby clusters to the present one, which are similarly gray and offer information about possible future steps.

Discussion

Indeed, while our study presents a promising framework that combines a recommender system with design space exploration, the true measure of its effectiveness lies in obtaining

feedback from designers. While there are several advantages to the proposed framework employing dimensionality reduction, clustering, and reinforcement learning-based recommender systems, there are also a number of limitations that need to be resolved in further studies.

- **Generalizability to Different Simulation Outcomes:** The current investigations focused on predicting the EnergyPlus outcome, specifically the yearly energy consumption per square meter for cooling, heating, and lighting. It is crucial to determine how well the proposed clustering-based performance prediction method would work in approximating other different types of performance outcomes, such as indoor air quality, daylighting, or thermal comfort.
- **Balancing Visual and Performance Outcome Differences while Clustering:** In this framework, clustering assumes that design options with different performance outcomes will also exhibit visual differences. To enhance the efficiency of the proposed framework, it is beneficial to cluster design options based on both visual and performance outcome differences.
- **Applicability to Different Building Types and Contexts:** The study explored a “mid-rise apartment” building, with randomly varying window sizes in different orientations. However, this approach may not directly apply to all types of apartment buildings or other building typologies.
- **Addressing Designer Subjectivity and Biases:** The mental state and subjective preferences of designers can influence the learning process of the reinforcement learning algorithm. Addressing individual biases and preferences to provide personalized and unbiased design recommendations requires further research and consideration.

The dashboard was carefully developed and made available online through the Google Cloud service, enabling global user engagement. While it offers valuable functionality, there is recognition of the need for improvements in usability and comprehensiveness. The main focus was on showcasing the potential of reinforcement learning-based recommender systems in design space exploration, with plans to refine and improve the dashboard based on user feedback and further research.

Conclusion

This study introduces a pioneering approach to exploring vast design spaces by integrating human and machine intelligence to generate high-performing and visually appealing design solutions. Unlike previous research relying on guided optimizations, our study employs a reinforcement learning-based recommender system that learns from designer interactions to suggest design options aligned with their preferences while minimizing annual energy consumption. To achieve effective recommendations, the algorithm needs to learn the performance outcomes of various design options. Recognizing the time-intensive nature of simulating all possibilities, we developed a framework that approximates performance outcomes by learning from a subset of the dataset. Remarkably, by simulating only 20% of the dataset, we achieved accurate predictions of the remaining design options’ performance.

We used powerful ML algorithms to rapidly and effectively explore broad design areas, focusing on yearly energy usage as a vital indicator due to its intricate connection with

numerous design aspects. This innovative approach leverages design computation and artificial intelligence to address an important problem in the field, resulting in fast and accurate predictions that showcase its sophistication and intelligence. Moreover, this process contributes to sustainability efforts by significantly reducing computation time and encourages widespread participation in design decision-making through the interactive design dashboard.

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